

8.2 Let us say that the set of nodes are ordered from 1 to K. For each node, there are no links going to a lower-numbered node. This means that for every node i , $pa_i \subseteq \{1, 2, \dots, (i - 1)\}$.

Similarly, for every node j in the set pa_i , $pa_j \subseteq \{1, 2, \dots, (j - 1)\}$ where $j < i$.

To have a directed cycle, at some point, we would need to return to node i . However, following successive parent links produces a strictly decreasing sequence of node numbers, making it impossible to return to the starting node. Hence, no directed cycle can exist.

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